FNB App of the Year Hackathon 2026: Comprehensive Project & Architecture Plan (v2.0 — Production-Hardened)

Track: Safer Communities and Gender-Based Violence Response
Event Date: Friday, 23 October 2026 (Registration Confirmation)
Lead Developer: Jayden Byron Maans (Lume Systems (Pty) Ltd)
Working Codename: Khusela / SafeLedger — *Camouflaged Offline-First Evidence Vault, Non-Blocking AI Voice Proxy & Official Form 06 (J480) Overlay Co-Pilot*

1. Executive Summary & Core Value Proposition

Khusela (SafeLedger) is a privacy-first, camouflaged Progressive Web App (PWA) and WhatsApp-integrated safety platform engineered for South African survivors of domestic and gender-based violence (GBV). Built for the FNB App of the Year Hackathon problem statement—*"How might technology help people find trusted support, document incidents safely, access emergency help and strengthen prevention without placing survivors at greater risk?"*—this **v2.0 Production-Hardened Plan** incorporates rigorous mobile browser, cryptographic, legal, and fault-tolerance guardrails so every technical claim stands up to scrutiny from senior engineering judges:

1. **Co-habitation Surveillance & Zero-Leak Camouflage:** Because **47.2% of reported rapes in South Africa occur inside the home of the victim or perpetrator** (SAnews), visible safety apps or loud panic buttons endanger survivors. Khusela operates inside a functional **Mzansi Smart-Budget & Grocery Calculator** on a single URL route (`/`) with no URL or title changes, salted PBKDF2 PIN verification (no hardcoded PINs in the JS bundle), in-memory `<canvas>` camera capture (bypassing the phone's photo gallery), and a **Screen Wake Lock** foreground duress mode.
2. **Realistic Cross-Browser Offline & Low-Data Resilience:** Financial abuse affects **1 in 8 South African women** and **16.3% of women with disabilities** (HSRC National GBV Study). Rather than relying on browser APIs that fail on iOS Safari, Khusela combines an **offline-first PWA App Shell** (syncing via `online` and `visibilitychange` events, with Service Worker `BackgroundSync` as a progressive enhancement on Android), an **in-production WhatsApp Social Bundle Bot** (Doksy architecture), and a clearly delineated **Native Capacitor/Operator USSD Roadmap** (`/api/v1/sos/ussd-simulate` in demo mode).
3. **Call-First, Enrich-Second AI Voice Proxy & Official Form 06 PDF Overlay:** During a duress trigger, Twilio places the outbound call **within 500ms** using a pre-synthesized **ElevenLabs** profile briefing (so emergency dispatch never depends on LLM latency), then asynchronously injects the **OpenAI Whisper + GPT-4o-mini** foreground room-audio summary if available. For legal protection under the **Domestic Violence Amendment Act 14 of 2021** (De Rebus), Khusela pairs the survivor's verbatim logs with

human-reviewed AI structured outputs and overlays them directly onto the official **Department of Justice Form 06 (J480) PDF** using `pypdf` and `ReportLab`.

2. South African Problem Validation & Market Evidence

Every architectural decision is anchored in verified South African crime, sociological, and legal data to substantiate the pitch to the FNB App of the Year Judges:

Problem Dimension	Verified South African Data & Legal Reality	Production-Hardened Architectural Response
Residential & Intimate Partner Danger	In Q4 2025/26 (Jan–Mar 2026), SAPS recorded **9,782 rapes**, with **47.2% (4,620 cases)** committed inside the victim's or perpetrator's residence by someone known to the victim, alongside **1,523 residential murders** (SAnews).	Single-route camouflaged calculator UI, hashed duress PIN, zero call-log traces, **no inbound callbacks to the survivor's phone**, and foreground mic capture while the calculator remains open on screen.
High Prevalence & Economic Control	The HSRC First South African National GBV Study found **33.1%** of women aged 18+ experienced physical violence, **>50%** experienced controlling behaviour, **25%** emotional abuse, and **12.5%** economic/financial abuse.	Offline-first PWA (`online`/app-open sync) requiring **0 MB data** at time of logging, paired with a low-data **WhatsApp Social Bundle Bot** and FNB Connect zero-rating model.
Intersecting Vulnerability (Disability & Age)	Women with disabilities experience **twice the rate of sexual violence (14.6% vs. 7.2%)** and **16.3% economic/grant exploitation**, while older township women face severe family financial and physical abuse (HSRC).	**ElevenLabs** multilingual audio guidance (English, Afrikaans, isiXhosa, isiZulu) for DoJ Form 08 (J506) legal rights and voice-first incident logging.
Digital Evidence & Court Form Friction	While the **Domestic Violence Amendment Act 14 of 2021** and **Cybercrimes Act 19 of 2020** enable electronic protection orders (De Rebus), the Department of Justice portal	Server-timestamped, tamper-evident after sync evidence vault + `pypdf`/`ReportLab` coordinate overlay directly onto the official **Form 06 (J480) PDF** with mandatory human review.

Problem Dimension	Verified South African Data & Legal Reality	Production-Hardened Architectural Response
	notes that forms remain static plain PDFs.	

3. Competitive Landscape & Gap Analysis

Existing South African solutions provide helplines or static directories, leaving a clear gap for an offline-capable, camouflaged evidence and legal preparation tool:

Existing Platform	Primary Capabilities	Key Limitations	How Khusela Complements & Advances the Ecosystem
Bright Sky SA (Vodacom Foundation)	Relationship risk questionnaires, abuse education, and geolocation directory of SAPS stations, hospitals, and NGOs.	Requires native app download; offers static directories rather than tamper-evident evidence syncing or protection order PDF population.	Runs as an install-free camouflaged PWA + WhatsApp bot; overlays verified survivor logs directly onto the official **DoJ Form 06 (J480) PDF**.
GBV Command Centre (`0800 428 428` / `*120*7867#`)	24/7 toll-free national call centre with social workers, "Please Call Me" USSD (`*120*7867#`), and SASL video calling.	Requires the survivor to speak on a voice call when an operator calls back—unsafe when the abuser is in the same room.	Places an **outbound-only ElevenLabs Voice Proxy call** from the cloud to opted-in trusted contacts/responders so the survivor's phone never rings or shows a call log.
TEARS Foundation (`08000 83277` / `*134*7355#`)	Free USSD support locator (`*134*7355#`), crisis intervention, and referral to Thuthuzela Care Centres (TCCs).	Focused on immediate helpline triage rather than longitudinal incident logging, escalation detection, or court form preparation.	Embeds verified TEARS and Thuthuzela Care Centre referrals directly into the post-incident workflow.

4. Technical Audit & 16 Production-Hardened Engineering Fixes

Following a pre-hackathon technical architecture review, all 16 browser, OS, cryptographic, legal, and latency risks have been resolved in the system design and pitch narrative:

#	Technical / Legal Risk Identified	Production-Hardened Engineering Fix in v2.0	MVP (PWA) vs. Roadmap Classification
1	**Background Audio Capture in a PWA:** Browsers (especially iOS Safari) block mic recording when the screen locks or the browser is backgrounded.	**Foreground Decoy Capture + Screen Wake Lock:** Audio capture runs strictly for 10–15s **while the calculator stays open in the foreground** using `navigator.wakeLock.request('screen')` so the screen does not lock while sitting on a table. Stated transparently in the pitch.	**Live in Friday MVP**
2	**Background Sync API Compatibility:** Service Worker `BackgroundSync` is unsupported on iOS Safari and inconsistent across some Android WebViews.	**Event-Driven Sync Primary:** Primary sync triggers on `window.addEventListener('online')`, `document.onvisibilitychange`, and when tapping "Update Grocery Prices" on the calculator. `SyncManager` is used only as a progressive enhancement where supported.	**Live in Friday MVP**
3	**USSD Trigger & Arbitrary Payloads:** Handset USSD requires an operator-issued shortcode, cannot carry arbitrary `*lat*lng#` payloads without an interactive menu, and PWAs can only open `tel:` links visibly.	**Simulated Operator Webhook + Roadmap:** In the PWA demo, USSD is demonstrated via `/api/v1/sos/ussd-simulate` (modeling a multi-step Africa's Talking / Clickatell operator menu session) and clearly presented as a **post-hackathon telecom integration roadmap item**.	**Simulated in MVP / Telecom Roadmap**
4	**Silent SMS Fallback in a PWA:** PWAs cannot send SMS silently (`sms:` opens the native messaging app).	**WhatsApp Bot Primary + Capacitor Native Wrapper Roadmap:** Low-data messaging in the MVP uses our server-side **Twilio WhatsApp Social Bundle Bot** (Doksy); silent background SMS via Android `SmsManager` is explicitly scoped to the **Capacitor native Android wrapper roadmap**.	**WhatsApp Live in MVP / Native SMS Roadmap**
5	**Reverse Call to Capture Room Audio:** Calling the survivor's	**Completely Dropped:** The system **never** places an inbound call to the	**Removed for Survivor Safety**

#	Technical / Legal Risk Identified	Production-Hardened Engineering Fix in v2.0	MVP (PWA) vs. Roadmap Classification
	phone back exposes them via ringing, call logs, and missed-call notifications, plus wiretap/consent liabilities.	survivor's handset during an emergency. All voice calls are strictly outbound from Twilio to opted-in responders/contacts.	
6	**Client-Side Crypto & 4-Digit PIN Brute-Force:** A 4-digit PIN has only 10,000 combinations, and IndexedDB is persistent disk storage, not "volatile RAM".	**Device-Bound Key Wrapping + PBKDF2 + 3-Strike Wipe:** Ciphertext at rest in IndexedDB is encrypted with `AES-256-GCM` using a key derived via **PBKDF2 (600,000 iterations, SHA-256)** combining the user's PIN **plus a high-entropy server-escrowed/device-bound secret token**, with strict rate limiting and an automatic **local IndexedDB wipe after 3 failed PIN attempts**. Plaintext exists only in JS memory while unlocked.	**Live in Friday MVP**
7	**Evidence Integrity Claims:** A client-side SHA-256 and device clock can be forged by the client and cannot be claimed as automatically "court-admissible".	**"Tamper-Evident After Sync" + Server/RFC 3161 Timestamp:** On sync, the FastAPI server hashes the payload, logs an **authoritative server-side UTC timestamp (and RFC 3161 Trusted Timestamping token)** in an append-only ledger, and frames the output as **"tamper-evident chain-of-custody from time of sync"** to assist courts under ECTA Section 15.	**Live in Friday MVP**
8	**Photo Handling & Gallery Leaks:** `<input type="file" capture>` saves photos to the native camera roll on many Android devices.	**In-App `<video>` Stream to Offscreen `<canvas>`:** Photos are captured inside the browser via `navigator.mediaDevices.getUserMedia({ video: { facingMode: 'environment' } })` rendered to an in-memory `<canvas>` (`canvas.toBlob('image/webp')`), which **never writes to the device's native photo gallery**.	**Live in Friday MVP**

#	Technical / Legal Risk Identified	Production-Hardened Engineering Fix in v2.0	MVP (PWA) vs. Roadmap Classification
9	**Opus 12 kbps Browser Inconsistency:** `MediaRecorder` Opus bitrates vary, and iOS Safari records `audio/mp4` (AAC) rather than Opus.	**Feature-Detected MIME Types + Server `ffmpeg` Normalization:** Client checks `MediaRecorder.isTypeSupported()` (`audio/webm;codecs=opus` on Chromium; `audio/mp4` with `audioBitsPerSecond: 16000` on Safari), and the FastAPI backend normalizes/transcodes uploaded audio via `ffmpeg`.	**Live in Friday MVP**
10	**Duress PIN Visibility & History Hygiene:** Hardcoded PINs (`9999`) are visible in JS bundles, and route changes leak in browser history.	**Salted Hash Comparison + Single-Route History Hygiene:** No plaintext PINs exist in code; PINs are verified against salted hashes (`timingSafeEqual`-equivalent constant-time flow) with identical UI timing. The app uses a single `/` URL (`history.replaceState`), never alters `document.title`, and uses a neutral `manifest.json` (*"Mzansi Smart-Budget"*).	**Live in Friday MVP**
11	**SOS Pipeline Latency & LLM Failure Risk:** Chaining `Audio → Whisper → GPT → ElevenLabs → Twilio` delays the emergency call and creates a single point of failure.	**Call-First, Enrich-Second Dead-Man's Pipeline:** Twilio dials immediately (`<500ms`) upon receiving the initial 250-byte ping using a **cached ElevenLabs profile audio file + template fallback** (zero LLM dependency). If the 10s foreground audio + Whisper/GPT summary completes within timeout, it is injected into the live call or sent via follow-up WhatsApp/SMS.	**Live in Friday MVP**
12	**AI Voice Proxy Abuse & False Alarms:** Unverified numbers could be spammed with terrifying automated distress calls.	**Double Opt-In Contacts, Rate Limits & Silent Cancel Window:** Trusted contacts must complete a WhatsApp/SMS opt-in confirmation when added; SOS calls are rate-limited per account with a full	**Live in Friday MVP**

#	Technical / Legal Risk Identified	Production-Hardened Engineering Fix in v2.0	MVP (PWA) vs. Roadmap Classification
		audit log; vault-triggered SOS includes a 5-second silent cancel bar.	
13	**LLM Hallucination Risk in Legal Forms:** LLMs must never invent dates, injuries, or legal claims in a court affidavit.	**Strict JSON Schema + Verbatim Preservation + Mandatory Review:** OpenAI runs at `temperature: 0` with Structured Outputs schema validation, displaying the **survivor's original verbatim transcript side-by-side** with the structured text for mandatory human review/editing prior to PDF export.	**Live in Friday MVP**
14	**Form 06 PDF Legal Layout Fidelity:** Recreating a statutory form from scratch risks rejection by clerks of the court.	**Coordinate Overlay on Official DoJ PDF (`pypdf` + `ReportLab`):** Instead of redrawing the form, the backend generates a transparent `ReportLab` text layer and merges it (`page.merge_page()`) directly onto the official **Department of Justice Form 06 (J480) PDF**.	**Live in Friday MVP**
15	**Server-Side Data Protection & Retention:** Storing raw PINs, persistent GPS trails, or indefinite audio increases breach liability.	**Per-User Envelope Encryption + Short Retention + Auditable "Quick Burn":** Evidence is encrypted at rest server-side with per-user keys; PINs and passive GPS are never logged; raw SOS audio has short configurable retention; **"Quick Burn"** permanently purges database rows, object storage, and application logs.	**Live in Friday MVP**
16	**Live Hackathon Demo Robustness:** Live cellular calls, iOS/Android quirks, or API hiccups can derail a 3-minute stage pitch.	**Whitelisted Demo Mode + Pre-Warmed SW + Recorded Fallback:** Demo mode restricts outbound Twilio calls strictly to whitelisted test numbers (e.g., judge/team phone), pre-warms the Service Worker on load, is tested across iOS Safari and Android	**Live in Friday MVP**

#	Technical / Legal Risk Identified	Production-Hardened Engineering Fix in v2.0	MVP (PWA) vs. Roadmap Classification
		Chrome, and includes an instant local audio fallback if venue Wi-Fi drops.	

5. Practical Offline & Low-Data Resilience Architecture (PWA MVP + Roadmap)

To address South Africa's data and connectivity constraints honestly and effectively, Khusela separates **what works today in a zero-install mobile browser (PWA + WhatsApp)** from **what requires a native wrapper or telecom partnership on the post-hackathon roadmap**:

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Connectivity Layer	Status	How It Works Technically	Survivor Experience
Layer 1: Offline-First PWA & Event-Driven Sync	**Live in Friday MVP**	Caches the calculator UI, risk questionnaire, Form 08 (J506) rights guide, and Thuthuzela Care Centre directory via Service Worker (`<400 KB`). Encrypts offline logs in IndexedDB (`AES-256-GCM`) and syncs automatically on `window.ononline` or next app open.	Opens and logs photos/voice notes with **0% signal and 0 MB data**; syncs and wipes local IndexedDB ciphertext automatically when next connected to Wi-Fi or mobile data.
Layer 2: 250-Byte Fast-Path SOS Burst	**Live in Friday MVP**	Sends a tiny 250-byte JSON packet (`user_id`, `lat`, `lng`, `timestamp`) over HTTPS before attempting any audio upload.	Works even on congested 2G/Edge connections; triggers the cloud **Twilio + ElevenLabs** call within 500ms without waiting for audio upload.
Layer 3: WhatsApp "Social Bundle" Bot (Doksy)	**Live in Friday MVP**	Uses the Twilio WhatsApp Business API so survivors who only have a R5 WhatsApp bundle (and 0 MB general web data) can message a disguised contact (*"Mzansi Daily Specials"*).	Lets survivors log voice notes/photos, receive ultra-compressed **ElevenLabs** audio guidance, and trigger an SOS to their trusted circle using only a WhatsApp bundle.

Connectivity Layer	Status	How It Works Technically	Survivor Experience
Layer 4: Capacitor Native SMS & Operator USSD (`*120*...#`) + FNB Connect Zero-Rating	**Simulated in MVP (`/sos/ussd-simulate`) / Phase 2 Roadmap**	Scopes an Android Capacitor wrapper (`SEND_SMS` permission for silent 160-byte SMS fallback) and an operator USSD menu gateway (modeled on GBV Command Centre *120*7867# and TEARS *134*7355#) plus **FNB Connect MVNO** IP zero-rating.	Demonstrates full product vision and telecom scalability to the FNB judges without making unrealistic browser claims.

6. Hardened System Architecture & Decoupled AI Pipeline

Component 1: Zero-Leak Camouflaged Frontend (React + Vite PWA)

- **Cover Persona (*Mzansi Smart-Budget & Grocery Calculator*):**
  - Web App Manifest (`manifest.json`), `<title>`, and home-screen icon are configured strictly as *"Mzansi Smart-Budget"*.
  - All views render inside a single root component at `/` using in-memory React state—zero URL path changes, zero `window.location.hash` changes, and `window.history.replaceState(null, '', '/')` on every state transition.
- **Constant-Time Salted PIN Routing & 3-Strike Shredder:**
  - During safe onboarding, the survivor sets a **Vault PIN** and a **Duress PIN**. Only salted PBKDF2 hashes (`600,000` iterations) are stored alongside a server-issued high-entropy device secret.
  - When `=` is pressed on the calculator, the input is hashed and compared in constant time:
    - **Normal Math Expression:** Computes and displays the grocery/VAT total.
    - **Duress PIN Match:** Displays a realistic budget total on the calculator (identical UI and timing to normal math), acquires a **Screen Wake Lock** (`navigator.wakeLock.request('screen')`) so the phone stays awake on the table, immediately fires the **250-byte Fast-Path SOS ping**, and records a 10–15 second foreground audio clip (`audio/webm;codecs=opus` or `audio/mp4` on iOS Safari).
    - **Vault PIN Match:** Decrypts IndexedDB blobs into volatile JS memory and renders the hidden Survivor Vault, with a 1-tap **"Quick Exit"** button that zeros out in-memory state and returns to the calculator in `<16ms`.
    - **3 Consecutive Failed PIN Attempts on Unlocked Vault:** Automatically shreds local IndexedDB keys and reverts purely to the calculator.
- **Gallery-Safe In-Memory Camera Capture:**

- Uses an inline `getUserMedia({ video: { facingMode: 'environment' } })` video element and captures still frames onto an offscreen `<canvas>` (`canvas.toBlob('image/webp', 0.8)`), ensuring evidence photos **never appear in the phone's native gallery or iCloud/Google Photos backup**.

Component 2: Non-Blocking "Call First, Enrich Second" SOS Pipeline (FastAPI + Twilio + ElevenLabs + OpenAI)

To guarantee that an emergency call is **never delayed or blocked** by audio upload times or LLM/TTS API outages, the backend (`/api/v1/sos/trigger`) uses a two-stage decoupled architecture:

1.

```
None
[Stage A: T+0.0s to T+0.5s — Immediate Dead-Man's Dispatch (Zero LLM
Dependency)]
Survivor Duress Trigger (250B JSON: user_id + GPS)
   └──> FastAPI verifies rate-limit & loads encrypted User Safety Profile
   └──> Twilio Voice API immediately dials Whitelisted / Opted-In Emergency
Contacts
   └──> Plays pre-cached ElevenLabs Profile Audio + Twilio <Say> Fallback with
live GPS link
        ("Urgent safety alert for [Name] at [GPS/Address]. Hold for live room
update or press 1 to acknowledge.")

[Stage B: T+1.0s to T+12.0s — Asynchronous Audio & AI Enrichment (With Timeout
Fallback)]
10s Foreground Audio Upload (WebM/Opus or MP4/AAC)
   └──> FastAPI normalizes via ffmpeg, computes SHA-256 + Server UTC Timestamp
   └──> OpenAI Whisper transcribes audio (3s timeout) -> GPT-4o-mini extracts
threat markers (2s timeout)
   └──> ElevenLabs Turbo v2.5 synthesizes 5s Threat Update clip (or falls back
to Twilio <Say> if ElevenLabs times out)
   └──> Injected into active Twilio call via Call Update webhook + sent via
WhatsApp/SMS with live tracking URL
```

2. **Anti-Abuse & False-Alarm Guardrails:**
   - Every emergency contact must complete a **WhatsApp/SMS Opt-In Verification** before being activated.
   - Outbound calls are strictly rate-limited (max 3 alerts per 15 minutes per user) and logged in an immutable audit table.
   - In **Judge Demo Mode** (`DEMO_MODE=true`), outbound calls can only dial numbers on the verified `DEMO_WHITELIST_NUMBERS` list (such as the judge's or presenter's phone on stage), with a pre-recorded local fallback clip ready if venue cell reception drops.

Component 3: Tamper-Evident Evidence Ledger & Official Form 06 (J480) PDF Overlay Engine

To respect South African legal standards under the **Domestic Violence Amendment Act 14 of 2021** and **Electronic Communications and Transactions Act (ECTA)** (De Rebus):

- **1. Tamper-Evident Server Timestamping (Not Client-Only Claims):**
  - When an offline or online evidence item (photo, screenshot, or voice note) syncs to `/api/v1/evidence/sync`, the server computes the **SHA-256 digest** of the raw file, binds it to an **authoritative server UTC timestamp** (and optional **RFC 3161 Trusted Timestamping Authority response**), and stores the file using **per-user envelope encryption (`AES-256-GCM`)**.
  - In the pitch and generated certificate, this is accurately presented as **"tamper-evident chain-of-custody verification from time of server sync"**, proving the file was not altered after upload.
- **2. Zero-Hallucination Legal Structuring with Mandatory Human Review:**
  - **OpenAI (`gpt-4o-mini`, `temperature: 0`, Structured Outputs JSON Schema)** classifies incidents across the 20 statutory categories in **Form 06 Section 6(a)** and drafts chronological entries for Section 4.
  - **Strict Anti-Hallucination Prompt & UI:** The model is instructed never to infer dates, weapons, or injuries not explicitly stated by the survivor. The UI displays the **survivor's original verbatim voice transcript side-by-side** with the AI-structured paragraph, requiring the survivor to review, edit, and check a confirmation box before export.
- **3. 1:1 Coordinate Overlay on Official DoJ Form 06 (J480) PDF (`pypdf` + `ReportLab`):**
  - Instead of recreating the court form from scratch, `/api/v1/legal/generate-form06` loads the official **Department of Justice Form 06 (J480) PDF** via `pypdf.PdfReader`, uses `ReportLab` to render the survivor's reviewed responses onto transparent coordinate overlays matching pages 1–12 (including confidential **Form 6A** and **Section 11 Index of Annexures**), and merges them via `page.merge_page()`.

7. Responsible Data Governance, POPIA Compliance & "Quick Burn"

- **Per-User Envelope Encryption at Rest:** Each user record has a unique Data Encryption Key (DEK) wrapped by the server Master Key (KEK), ensuring database dumps cannot expose survivor narratives.
- **Zero Sensitive Telemetry in Logs:** Application and proxy logs strip PIN hashes, GPS coordinates, phone numbers, and transcript bodies.

Official Judging Pillar (FAQs)	How Khusela v2.0 Wins Technical & Impact Scrutiny
1. Product	Deployed, bug-free React PWA + FastAPI backend with realistic browser mechanics (Wake Lock foreground audio, `<canvas>` gallery-free camera, `pypdf` official Form 06 overlay) and remote judge login.
2. Intelligence	Longitudinal escalation risk scoring across logged incidents + real-time Whisper/GPT threat extraction that enriches active emergency calls without becoming a blocking single point of failure.
3. Creativity	Combines a constant-time camouflaged calculator, a **Call-First/Enrich-Second ElevenLabs Voice Proxy**, and a 1:1 overlay engine for official South African Domestic Violence Court forms.
4. Usability	Works on both iOS Safari and Android Chrome with zero app-store install, supports voice-first logging in English, Afrikaans, isiXhosa, and isiZulu, and provides side-by-side human review for court forms.
5. Impact	Directly addresses South Africa's crisis where **47.2% of rapes occur in the home** (SAnews) and turns static Department of Justice PDFs into accessible digital protection orders.
6. Sustainability	Low per-incident compute cost; clear roadmap for **FNB Connect MVNO zero-rating**, Capacitor native wrapper, and CSR sponsorship via the GBVF Response Fund.
7. Presentation	Transparent, engineering-first pitch that proactively explains *why* the app captures audio in the foreground with Wake Lock, *why* Twilio dials before the LLM runs, and *how* evidence integrity is verified at sync.

9. Verified References & Source Links

- FNB App of the Year Hackathon 2026 — Official Problem Statements
- FNB App of the Year — App Categories 2026
- FNB App of the Year — Frequently Asked Questions & Judging Criteria
- FNB App of the Year — Official Terms & Conditions
- SAnews — Cachalia: Gender-Based Violence Persists Behind Closed Doors (Q4 2025/26 Crime Stats)
- Human Sciences Research Council (HSRC) — Violence Against Women in South Africa: Intersecting Vulnerabilities (First National GBV Study)

- De Rebus — From Home to Handset: Confronting Tech-Facilitated Domestic Violence in South Africa (Domestic Violence Amendment Act 14 of 2021 & Cybercrimes Act 19 of 2020)
- Department of Justice and Constitutional Development — Domestic Violence Official Forms Portal
- Department of Justice — Official Form 06 (J480): Application for Protection Order PDF
- Gender-Based Violence Command Centre (0800 428 428 / *120*7867#)
- Vodacom Foundation / GBV Command Centre — Bright Sky SA Overview
- TEARS Foundation — Support Locator (*134*7355#) & Services
- South African Government — Thuthuzela Care Centres (TCCs)
- GBVF Response Fund — Get Support Directory

10. Comprehensive 5-Pillar Feasibility Study (TELOS Framework)

To demonstrate commercial, technical, legal, and operational viability for the FNB App of the Year Judging Panel (specifically addressing the **Product**, **Sustainability**, and **Impact** criteria), this feasibility study evaluates **Khusela (SafeLedger)** across five dimensions: **Technical, Economic, Legal/Regulatory, Operational, and Schedule (TELOS)**.

10.1 Technical Feasibility

- **Browser & OS Compatibility (Verified):** All MVP features rely on standard W3C Web APIs supported across Android Chrome and iOS Safari without requiring app-store approval:
    - **Offline App Shell & Storage:** Service Worker Cache API (`<400 KB` bundle) and IndexedDB with `window.crypto.subtle` (`AES-256-GCM` + `PBKDF2` at 600,000 iterations).
    - **Foreground Audio & Screen Persistence:** `MediaRecorder` with dynamic MIME detection (`audio/webm;codecs=opus` on Chromium; `audio/mp4` AAC at `16 kbps` on iOS Safari) paired with the W3C Screen Wake Lock API (`navigator.wakeLock.request('screen')`) so the decoy calculator remains awake on a table during a 10–15 second foreground duress capture.
    - **Gallery-Free Camera Capture:** `navigator.mediaDevices.getUserMedia({ video: true })` piped to an in-memory HTML5 `<canvas>` (`canvas.toBlob('image/webp')`), preventing writes to the phone's camera roll.
- **Backend & AI Latency Feasibility (Verified):**
    - By decoupling the emergency call from the AI transcription pipeline (**Call-First, Enrich-Second**), the initial outbound call via Twilio Programmable Voice triggers within `<500ms` of the 250-byte JSON ping using pre-cached ElevenLabs Flash/Turbo v2.5 audio (75ms–300ms synthesis latency).
    - Asynchronous room-audio transcription via OpenAI GPT-4o Mini Transcribe / Whisper and threat structuring via GPT-4o Mini complete in `2.5s–4.0s` for a

10-second clip, injecting a live call update without risking a dropped dispatch if an LLM endpoint times out.

10.2 Economic & Unit-Economics Feasibility (API Cost Breakdown)

A critical question from FNB judges under the **Sustainability** pillar is whether an AI-powered safety platform is financially sustainable at scale in South Africa. Using verified API pricing (converted at `~$1 = R18.00`), the marginal cost per incident and per active survivor is fractionally small:

Workflow / Transaction	Underlying API / Infrastructure	Verified Unit Rate	Usage per Event	Cost per Event (USD)	Cost per Event (ZAR)
1. Evidence Voice Note Logging (30s clip + AI Structuring)	OpenAI Transcription (gpt-4o-mini-transcribe / Whisper) + GPT-4o Mini	`$0.003–$0.006 / min` audio + `$0.15 / 1M` input tokens	0.5 min audio + ~800 tokens	`~$0.0018`	**~R0.03**
2. Multilingual Audio Rights Guide Playback	ElevenLabs Flash / Turbo v2.5 API	`0.5 credits / char` (`~$0.05 / 1,000 chars` at scale, WaveSpeedAI / ElevenLabs)	Pre-cached static .ogg per language (amortized across all users)	`~$0.0000` (Cached)	**R0.00 (Cached)**
3. Silent Duress SOS (Outbound AI Voice Call + Live Audio Triage)	Twilio Programmable Voice SA + ElevenLabs Turbo v2.5 + OpenAI Whisper	Twilio Voice (`~$0.03–$0.05 / min`) + 250 chars dynamic TTS (`~$0.012`) + 15s Whisper (`~$0.0015`)	1-minute emergency call + 15s room audio triage	`~$0.0535`	**~R0.96**
4. WhatsApp "Social Bundle" Triage / Alert	Twilio WhatsApp Messaging API	`$0.005 / message` + Meta utility template fee	2 messages per session	`~$0.0150`	**~R0.27**
5. Official DoJ Form 06 (J480) PDF Generation	GPT-4o Mini Structured Outputs + Python pypdf/ReportLab	`$0.15 / 1M` input, `$0.60 / 1M` output tokens (local CPU PDF merge)	~3,000 input + ~1,500 output tokens	`~$0.0014`	**~R0.03**

10.2.1 How Khusela Pays for Itself: The "Never Charge the Survivor" B2B2C Cross-Subsidy Engine

In a domestic violence context, charging the survivor directly is both ethically unacceptable and physically dangerous—because **12.5% (1 in 8) of South African women** experience economic/financial abuse where an abuser monitors bank statements and airtime (HSRC National GBV Study). Any visible subscription charge on a bank statement would expose the survivor.

Instead, following the proven South African B2B2C embedded-safety playbook pioneered by platforms like AURA for Financial Services, Khusela remains **100% free and bank-statement-invisible for every survivor**, while generating high-margin commercial revenue from Day 1 across four institutional streams whose costs are directly reduced by our automation:

Revenue Stream	Target B2B / Institutional Customer	Commercial Value Delivered (Why They Pay)	Pricing Model (ZAR)	Gross Margin & Cross-Subsidy Impact
Stream 1: Legal Insurer & Banking Value-Added Service (VAS) Intake API	**FNB Law on Call**, FNB Life, LegalWise, Scorpion, Clientèle Legal	Currently, when a policyholder calls a legal insurer for a Domestic Violence Protection Order, a paralegal spends **2–3 billable hours (R1,500–R3,000+ in staff cost)** interviewing the client and manually typing up the 12-page **DoJ Form 06 (J480)**. Khusela's API delivers a pre-structured, human-verified Form 06 + SHA-256 Annexure Bundle at a compute cost of **R0.03**, cutting insurer intake costs by **>85%**.	**R150 per completed Form 06 intake packet** OR **R2.00 per policyholder/month** bundled into legal cover	**~98% gross margin**; 1 insurer contract (50,000 lives) generates **R100,000/month** in recurring revenue.
Stream 2: Employer Duty-of-Care & Campus Safety SaaS (PEPM)	Cape Town & national BPO call centres, retail groups, mines, hospitals, and universities (following TEARS	Helps employers comply with South Africa's **Employment Equity Act Harassment Code** and protect night-shift/commuting staff. Provides anonymized ESG/HR	**R4.00 per employee/student per month (PEPM)** (Blended active case cost at 3% monthly utilization =	**~99% gross margin**; 1 mid-sized employer/campus (5,000 seats = R20,000/mo) funds **14,000+ full township survivor cases**.

Revenue Stream	Target B2B / Institutional Customer	Commercial Value Delivered (Why They Pay)	Pricing Model (ZAR)	Gross Margin & Cross-Subsidy Impact
	Workplace GBV programmes)	safety heatmaps without exposing individual identities.	**R0.04/seat/month**)	
Stream 3: Private Security & Armed Response "Silent Voice Proxy" Licensing	AURA, Fidelity ADT, Armed Response control rooms, and neighbourhood watches	Standard panic buttons trigger high false-alarm dispatch costs (`R400+` per false truck roll) and fail when a victim cannot speak on a verification call. Khusela's **Call-First + Whisper/GPT Room Triage** gives control rooms verified acoustic threat context without ringing the victim's phone.	**R1.50 per active subscriber/month** API licensing fee + **R5.00 per verified SOS dispatch**	Eliminates costly false dispatches for security companies while funding Khusela's Twilio/ElevenLabs cloud infrastructure.
Stream 4: Zero-Rated CSR & Public-Benefit Sponsored Tiers	**FirstRand Foundation / FNB Connect**, GBVF Response Fund, Telkom/Vodacom Foundations	Provides measurable, audited ESG and B-BBEE Socio-Economic Development (SED) impact: sponsors fund a pre-paid pool of survivor cases with real-time anonymized dashboards showing exact protection orders generated and SOS dispatches completed.	**R150,000 annual CSR sponsorship packages** + **FNB Connect IP Zero-Rating**	Covers **106,000+ complete survivor case lifecycles** per R150k sponsorship at **R1.41/case**.

- **Day-1 Break-Even Math for Judges:**
  - Fixed monthly cloud overhead (Railway/Cloudflare hosting, domain, Twilio SA number rental at `$1.50/mo` per Twilio SA Pricing, and baseline DB): **~R650/month**.
  - Variable cost per full survivor case lifecycle (5 voice logs + 1 WhatsApp alert + 1 SOS call + 1 Form 06 PDF): **R1.41**.
  - **Break-even point:** Signing **just 1 small BPO employer or campus cohort of 200 seats at R4/month (R800/mo)** or processing **5 legal-insurer Form 06 intakes at R150 each (R750/mo)** makes the entire platform **100% cash-flow positive** while keeping the app completely free for every survivor in South Africa.

10.3 Legal & Regulatory Feasibility (South African Statutory Compliance)

- **Domestic Violence Amendment Act 14 of 2021 Compliance:** Section 4(1) and the 2023 Domestic Violence Regulations explicitly authorize electronic submission of **Form 06 (J480)** to Domestic Violence Courts (Department of Justice Forms Portal). Using `pypdf` + `ReportLab` to overlay text onto the official government PDF preserves 100% statutory layout compliance, while keeping the survivor's verbatim transcript alongside the AI-structured text with mandatory human sign-off prevents unauthorized legal practice or AI fabrication.
- **ECTA Section 15 & Cybercrimes Act 19 of 2020 Evidentiary Weight:** Under Section 15 of the Electronic Communications and Transactions Act (ECTA), the evidential weight of a data message depends on the reliability of the manner in which it was generated, stored, and maintained (De Rebus). By recording an authoritative server-side UTC timestamp and SHA-256 digest upon sync (and framing the certificate strictly as **"tamper-evident from time of server sync"**), Khusela strengthens evidentiary integrity without making legally invalid guarantees of automatic court admissibility.
- **POPIA & RICA / Telecommunications Compliance:**
  - Dropping the "reverse call listen-in" eliminates unlawful interception risks under RICA.
  - Requiring double opt-in for Trusted Emergency Contacts and enforcing per-user envelope encryption (`AES-256-GCM`) with a cascading **"Quick Burn"** right-to-erasure satisfies POPIA Conditions 7 (Security Safeguards) and 8 (Data Subject Participation).

10.4 Operational & Schedule Feasibility (Hackathon Execution Window)

Because the team already holds working production code and API credits across **FastAPI, React + Vite PWA, Twilio, ElevenLabs, OpenAI, and ReportLab** (Jayden_Maans_CV-v2.pdf), the entire MVP is scoped to build and deploy cleanly within the hackathon window:

Build Milestone	Allocated Window	Deliverables & Verification Gate	Risk Mitigation
Milestone 1: Camouflaged PWA & Crypto State Machine	Hours 1–6	Single-route `/` Mzansi Smart-Budget calculator, salted PBKDF2 PIN routing (`Normal`, `Vault`, `Duress`), Screen Wake Lock, and `<canvas>` gallery-free photo capture.	Uses standard Web Crypto and Canvas APIs; zero external UI library dependencies.
Milestone 2: Decoupled FastAPI SOS & ElevenLabs Voice Proxy	Hours 7–14	`/api/v1/sos/trigger` (250B fast-path Twilio call + cached ElevenLabs audio) + async Whisper/GPT-4o-mini room	Pre-generates the fallback ElevenLabs audio file so the call succeeds even if live TTS/LLM is slow.

Build Milestone	Allocated Window	Deliverables & Verification Gate	Risk Mitigation
		summary update + `DEMO_MODE` number whitelist.	
Milestone 3: Evidence Ledger & Form 06 PDF Overlay	Hours 15–22	`/api/v1/evidence/sync` (SHA-256 + server UTC timestamp), side-by-side human review UI, and pypdf + ReportLab overlay onto **DoJ Form 06 (J480)**.	Overlays key high-impact pages (Pages 1, 3, 5, 6, and Annexure Certificate) first before mapping secondary fields.
Milestone 4: Cross-Device QA, Deployment & Pitch Video	Hours 23–30	Railway/Cloudflare deployment, iOS Safari + Android Chrome physical device testing, remote judge login credentials, and 3-minute live demo recording.	Keeps a recorded local backup of the Twilio/ElevenLabs phone call in case of stage cellular interference.

11. UI/UX Mockups & Interactive Wireframe Specifications

The application UX is structured across three single-route (`/`) interface states so there are zero URL or browser history leaks:

11.1 Screen 1 Specification: Camouflaged Decoy ("Mzansi Smart-Budget") & Silent Duress Trigger

- **Visual & Security Design:** Presents as a South African household grocery and VAT calculator showing zero-rated staples (Bread R18.50, Maize Meal R65.00) vs. 15% VAT items (Cooking Oil R42.00) in Rand (`R`), with an **"Offline Ready"** status pill confirming the Service Worker cache is active.
- **Zero-Trace Interaction Flow:** Typing the user's custom Duress PIN (e.g., `9999`) and pressing `=` keeps the grocery calculator completely unchanged on screen, acquires a **Screen Wake Lock**, fires the 250-byte Fast-Path SOS packet, and records 10–15 seconds of foreground audio.
- **Direct Drive Asset:** View High-Resolution Screen 1 Mockup in Google Drive (Asset ID: 1jC6c7pw22TL-IcPfdSJc16g2F6zzs2RW)
- **Note for Pitch Deck / Printing:** In Google Docs, you can also embed this image directly inline via Insert > Image > Google Drive and selecting "Khusela Mockup - Screen 1 (Mzansi Smart-Budget Decoy)".

11.2 Screen 2 Specification: Hidden Survivor Vault ("Khusela SafeLedger Vault")

- **Instant Panic-Exit Header:** Features a full-width sticky coral-red **"QUICK EXIT TO CALCULATOR"** bar at the very top of the viewport (paired with a hardware shake listener) that wipes decrypted JS state and snaps back to Screen 1 in `<16ms`.
- **Gallery-Free Evidence Capture Card:** Provides 1-tap buttons for **"Record Foreground Voice Note"** and **"Canvas Photo"** (capturing directly from `<video>` to `<canvas>` without touching the phone's camera roll) alongside a green **"Tamper-Evident Server Synced (SHA-256)"** verification badge.
- **AI Escalation Risk Analysis & Audio Rights Companion:** Displays the longitudinal **AI Escalation Risk Gauge** (`High Risk 78%` with detected statutory tags such as `Economic Control` and `Verbal Threats` to satisfy the FNB **Intelligence** criterion) and an **ElevenLabs Audio Legal Rights Guide** player in *English*, *Afrikaans*, *isiXhosa*, and *isiZulu*.
- **Direct Drive Asset:** View High-Resolution Screen 2 Mockup in Google Drive (Asset ID: 1_VPKXIQFPnBut-2Fec-867MPjne3GJFd)
- **Note for Pitch Deck / Printing:** In Google Docs, you can also embed this image directly inline via Insert > Image > Google Drive and selecting "Khusela Mockup - Screen 2 (Hidden Survivor Vault)".

11.3 Screen 3 Specification: Human-in-the-Loop Legal Review & DoJ Form 06 (J480) PDF Overlay

- **Side-by-Side Verbatim vs. AI Review (Left Panel):** Displays **"1. Verbatim Voice Transcript (Original)"** directly above **"2. AI Structured Statement (Section 4)"** with an explicit confirmation toggle (**"I confirm facts and dates are accurate"**) that must be enabled by the survivor before the **"Generate Form 06 (J480) PDF"** button activates.
- **1:1 Official Form 06 (J480) Overlay & Annexure Certificate (Right Panel):** Shows the generated PDF where `pypdf` and `ReportLab` overlay the survivor's verified statement onto the official **Department of Justice Form 06 (J480)** template alongside a cryptographic **SHA-256 Annexure Certificate** and the toggleable **Judge Demo Mode** bar.
- **Direct Drive Asset:** View High-Resolution Screen 3 Mockup in Google Drive (Asset ID: 1vS0kovzX5JHUdBcHhcM0KwZu1yuSi0y-)
- **Note for Pitch Deck / Printing:** In Google Docs, you can also embed this image directly inline via Insert > Image > Google Drive and selecting "Khusela Mockup - Screen 3 (Form 06 Review & PDF Overlay)".